

Chenyu Wang

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Research Interests

Artificial Intelligence, Computer Systems and Architecture

Education

Princeton University

Princeton, NJ

M.S.E., Department of Electrical and Computer Engineering

Sep. 2023 - May 2025

- Thesis Title: Optimizing Efficiency of Cognitive Agents with Improved Performance
- Thesis Advisor: Tom Griffiths

Tsinghua University

Beijing, China

B.E., Department of Electronic Engineering

Aug. 2019 - Jun. 2023

- GPA: 3.93/4.0, Ranking: 11/260+
- Graduated as Outstanding Graduate (Top 5%)
- Advisor: Prof. Yu Wang

Work Experience

Harvard University

Cambridge, MA

Research Fellow under Prof. Vijay Janapa Reddi

Jun. 2025 - Present

ML for Computer Systems and Architecture:

- **Foundation Models for Computer Systems and Architecture** (Lead Researcher): Driving the development of domain-specific foundation models for architecture reasoning. Training BERT-based embedding models for architectural knowledge representation and continue-pretraining GPT models for architecture-specific language understanding. Managing end-to-end model training pipeline including data curation, training infrastructure, and evaluation.
- **QUARCH Benchmark**: Co-developed QUARCH (submitted to ICLR 2026), a comprehensive benchmark with 2,671 expert-validated questions for evaluating LLM reasoning in computer systems and architecture.
- **Teaching Fellow for CS249r** (Agentic AI for Computer Systems Design): Gained systematic exposure to ML for computer systems and architecture through weekly seminars with leading researchers in the field. Co-authoring a comprehensive survey on AI for computer systems and architecture (expected 2026).
- Mentoring undergraduate intern on ML for architecture mapping project.

Reasoning and Multi-Agent AI (with Prof. Yilun Du):

- Led development of **SLM-MUX** (submitted to ICLR 2026), a framework for enhancing LLM compositional reasoning through multi-model orchestration. Core insight: LLM generation is constrained by pretraining distribution; by composing multiple within-distribution calls from small language models, we can surpass the performance of 72B-parameter large models.

Publications

* implies equal contribution.

System-Focused Publications

- Zishen Wan, Hanchen Yang, Jiayi Qian, Ritik Raj, Joongun Park, **Chenyu Wang**, Arijit Raychowdhury, and Tushar Krishna. "Compositional AI Beyond LLMs: System Implications of Neuro-Symbolic-Probabilistic Architectures", accepted by *Architectural Support for Programming Languages and Operating Systems* (ASPLOS'26 - Spring cycle), 2026.
- **Chenyu Wang***, Zhen Dong*, Daquan Zhou*, Zhenhua Zhu, Yu Wang, Jiashi Feng, and Kurt Keutzer. "[Efficient Processing-in-Memory Accelerators based on Epitome](#)", accepted by *Design Automation Conference* (DAC), 2024.
- Hanbo Sun, Zhenhua Zhu, **Chenyu Wang**, Xuefei Ning, Guohao Dai, Huazhong Yang, and Yu Wang. "[Gibbon: Efficient Co-Exploration of NN Model and Processing-in-Memory Architecture](#)", in *IEEE Transactions on Computer-Aided Design* (TCAD), 2023.
- Hanbo Sun*, **Chenyu Wang***, Zhenhua Zhu, Xuefei Ning, Guohao Dai, Huazhong Yang, and Yu Wang. "[Gibbon: Efficient Co-Exploration of NN Model and Processing-in-Memory Architecture](#)", accepted by *Design, Automation and Test in Europe Conference* (DATE), 2022.

Machine Learning-Focused Publications

- Shvetank Prakash, Andrew Cheng, Arya Tschand, Mark Mazumder, Varun Gohil, Jeffrey Ma, Jason Yik, Zishen Wan, Jessica Quaye, Elisavet Lydia Albanaki, Avinash Kumar, Chandrashis Mazumdar, Tuhin Khare, Alexander Ingare, Ikechukwu Uchendu, Radhika Ghosal, Abhishek Tyagi, **Chenyu Wang**, et al. "[QUARCH: A Benchmark for Evaluating LLM Reasoning in Computer Architecture](#)," submitted to *International Conference on Learning Representations* (ICLR), 2026. Under review.
- **Chenyu Wang***, Zishen Wan*, Hao Kang, Zhiqiang Xie, Tushar Krishna, Vijay Janapa Reddi, Yilun Du. "[SLM-MUX: Orchestrating Small Language Models for Reasoning](#)," submitted to *International Conference on Learning Representations* (ICLR), 2026. Under review.
- **Chenyu Wang**, Yihan Wang. "[Evaluating Zero-Shot Long-Context LLM Compression](#)", preprint available on *arxiv*, 2024.
- Yi Cai*, **Chenyu Wang***, Zixuan Zhou, Xuefei Ning, Huazhong Yang, and Yu Wang. "[Deep-Guise: Learning to Disguise Neural Architectures for Impeding Adversarial Transfer Attacks](#)", preprint available on *arxiv*, 2023.

Honors & Awards

Aug. 2023 – May. 2025	Fellowship: "Gordon Wu Fellowship"
Jun. 2023	Honorary Title: "Outstanding Graduate of Tsinghua University"
Jun. 2023	Honorary Title: "Outstanding Graduate of Beijing"
Jun. 2023	Award: "Outstanding Bachelor Thesis Award"
Jun. 2022	Award: "Academic Excellence Award at Tsinghua University"
Jun. 2022	Honorary Title: "Spark Innovative Talent Cultivation Program"
Jun. 2021	Award: "Academic Excellence Award at Tsinghua University"
Nov. 2020	Contest: First Prize in "Chinese Undergraduate Students' Physics Competition"
Jun. 2020	Award: "Academic Excellence Award at Tsinghua University"
Apr. 2018	Contest: Silver Medal in "35th Chinese Physics Olympiad"

Teaching Experience

Teaching Fellow, CS249r: Agentic AI for Computer Systems Design Fall 2025, Harvard University

Teaching Assistant, SML 312: Research Projects in Data Science Fall 2024 - Spring 2025 (2 semesters), Princeton University